LP Modeling

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July 15, 2026

Lookup

20ns 40ns

In the following we will use capital letters for constants and bold letters for variables.

Runtime

constant	meaning
${T_R}^{L_i}$	time to check presence of a bit in vector R of level i
${T_{SL}}^{L_i}$	time to lookup a k-mer in vector S on level i
variable	
T_H	time to lookup a k-mer on level i
variable	
$\mathbf{p_K}^{L_i}$	probability of a kmer belonging to level i

As a first approach we model the runtime complexity for only one level. In this case the lookup of a query with minimizer m of frequency f(m) would consist of checking the bitvector R in order to determin whether $1 \leq f(m) \leq threshold$. This will be the case for $\mathbf{p_K}$ of the k-mers, which will then be followed by a lookup in S. If m does not appear at all or has a $f(m) > threshold$, a lookup in the hashtable will follow. The lookup time for one level can thus be modeled as:

$$T_R + {T_{SL}}^{L_1} * \mathbf{p_K} + T_H * (1 - \mathbf{p_K})$$

minimize: nicht linear!

$$\begin{aligned} {T_R}^{L_1} \quad & + \mathbf{p_K}^{L_1} * {T_{SL}}^{L_1} \\ & + (1 - \mathbf{p_K}^{L_1}) * \Big({T_R}^{L_2} \quad + \mathbf{p_K}^{L_2} * {T_{SL}}^{L_2} \\ & \qquad + (1 - \mathbf{p_K}^{L_2}) * \Big({T_R}^{L_3} \quad + \mathbf{p_K}^{L_3} * {T_{SL}}^{L_3} \\ & \qquad\qquad + (1 - \mathbf{p_K}^{L_3}) * T_H \Big) \Big) \end{aligned}$$

TODO: linearise

variable	meaning
$\mathbf{p_K}^{L_i}$	probability of a kmer belonging to level i
${T_R}^{L_i}$	time to check presence of a bit in vector R of level i
$t_l^{L_i}$	time to lookup a k-mer on level i

for:

$$\mathbf{p}_{\mathbf{K}}^{L_i} = \sum_{m \in M} \left(\mathbf{sm}_m^{L_i} \sum_{t \in T} P_{K-empiric}^{L_i}(m,t) * \mathbf{st}_t^{L_i} \right)$$

$$\mathbf{p}_{\mathbf{K}}^{L_i} - \sum_{m \in M} \left(\mathbf{sm}_m^{L_i} \sum_{t \in T} P_{K-empiric}^{L_i}(m,t) * \mathbf{st}_t^{L_i} \right) = 0$$

TODO: linearise

variable	meaning
$\mathbf{sm}_m^{L_i}$	binary selector for minimizer length m in level i
$\mathbf{st}_t^{L_i}$	binary selector for threshold of t in level i
$P_{K-empiric}^{L_i}(m,t)$	empirically estimated value for a kmer with a minimizer of length m to have a frequency less than t
M	Set of all minimizer length to be considered
T	Set of thresholds to be considered

selectors for m and t must be binary amd mutually exclusive:

$$\mathbf{sm}_m^{L_i} \in 0,1 \mid \forall m \in M$$

$$\mathbf{st}_t^{L_i} \in 0,1 \mid \forall t \in T$$

$$\sum_{m \in M} \mathbf{sm}_m^{L_i} = 1$$

$$\sum_{t \in T} \mathbf{st}_t^{L_i} = 1$$

Space Consumption

Using Elias Fano Coding a Bitvector of length U with n bits set to 1 can be encoded using $n\lceil log\frac{U}{n}\rceil + 3n$.

Let $noam$ be the number of appearing minimizers (minimizers with a frequency greater than zero).

Level 1

R - Presence of k-mers:
probably not relevant R^{L_1} has length $4^{sm_m^{L_1}}$ and contains $p_M^{L_1} * noam$ many 1s.

$$\begin{aligned} s_R^{L_1} &= p_M^{L_1} * noam\lceil log\frac{4^{sm_m^{L_1}}}{p_M^{L_1} * noam}\rceil + 3 * p_M^{L_1} * noam \\ &= p_M^{L_1} * noam\left(\lceil log\frac{4^{sm_m^{L_1}}}{p_M^{L_1} * noam}\rceil + 3\right) \end{aligned}$$

S - Offset Boundaries:
S is a bitvector of length $|O| + 1$ and is **probably not relevant** for the optimization of m and t.

O - Offsets:
O^{L_1} contains one entry of size $log|T|$ for the offset of every minimizer with frequency $\leq t^{L_1}$ in T.

$$|O| = log_2|T| * \sum_{m=1}^{t^{L_1}} m * (\text{number of minimizers with frequenzy m})$$

Every minimizer of length m covers an average of $\frac{k-m+2}{2}$ consecutive k-mers.

$$bpk_O^{L_1} = \frac{2}{k - m^{L_1} + 2} * p_K^{L_1} log|T|$$